

Dingyuan Huang

Robotics Systems · Computer Vision · Physical AI

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[Google Scholar](#)

RESEARCH & ENGINEERING

DreamX Lab, University of Florida — Research Assistant

APR 2025 - PRESENT

Build end-to-end real-robot systems across UR5, WidowX AI, Piper, and humanoids, integrating RGB-D perception, ROS/Python control, manipulation, sensor feedback, and hardware validation.

Orthodontic Assistant Robot, UF — Robotics Researcher

2025 - PRESENT

Built a perception-to-action pipeline with RealSense D405/D455 for 3D localization, tweezer-based grasping, automated adhesive dispensing, XYZ correction, post-action verification, and clinician handoff.

Guoxuan New Energy, Hefei, China — Engineering Intern

MAR 2024 - MAY 2024

Optimized BMS control-board circuits in AutoCAD and modeled battery performance across loads in MATLAB for control-strategy evaluation.

EDUCATION

University of Florida, Gainesville, FL — Ph.D., Civil & Coastal Engineering

2026 - PRESENT

University of Florida, Gainesville, FL — M.S., Electrical & Computer Engineering

2024 - 2025

Central South University, China — B.E., Intelligent Manufacturing

2020 - 2024

SELECTED RESEARCH

Humanoid Robot Learning

Co-author, JIC 2026 & HOIST 2026 — VLA, imitation learning, RL.

SKILLS

Programming — Python, C/C++, ROS/ROS 2, Linux

Perception — OpenCV, YOLO, PyTorch, RGB-D / 3D vision

Robotics — manipulation, grasping, motion planning, visual servoing

Robot Learning — VLA, reinforcement learning, imitation learning

PLATFORMS

UR5, WidowX AI, Piper, humanoid platforms

RealSense D405/D455, Robotiq grippers/FT sensing, Jetson AGX Orin

LANGUAGES

Chinese (Native) · English (Professional)